

Internship Project Report

Submitted By

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Internship: Cybersecurity Internship

Repository: Internship-Projects

Project 1: Web Application Vulnerability Scanner

Introduction

The Web Application Vulnerability Scanner is a Python-based security tool developed to identify common vulnerabilities in web applications. It automates the scanning process and generates reports that help developers improve the security of their websites.

Abstract

The project scans websites for security vulnerabilities such as SQL Injection, Cross-Site Scripting (XSS), and missing security headers. It also stores scan history and generates PDF and CSV reports for future analysis. The application provides an easy-to-use web interface for performing vulnerability assessments.

Tools and Technologies Used

- Python
- Flask
- SQLite
- HTML
- CSS
- JavaScript
- Requests
- BeautifulSoup

Steps Involved in Building the Project

1. Designed the project architecture.

2. Developed the Flask web application.
3. Implemented website crawling.
4. Added SQL Injection detection.
5. Implemented XSS vulnerability scanning.
6. Added Security Header analysis.
7. Stored scan history using SQLite.
8. Generated PDF and CSV reports.
9. Designed the user interface.
10. Tested and debugged the application.

Conclusion

The Web Application Vulnerability Scanner successfully automates website security testing and provides detailed vulnerability reports. It helps developers identify and fix common security issues, improving overall web application security.

Project 2: Secure Image Steganography Tool

Introduction

The Secure Image Steganography Tool is a Python application developed to securely hide and retrieve secret messages inside digital images using steganography combined with encryption techniques.

Abstract

This project provides secure communication by embedding encrypted messages into images. The hidden information is invisible to users and can only be extracted using the correct process. The application includes a graphical interface for hiding and extracting messages.

Tools and Technologies Used

- Python
- OpenCV
- Cryptography
- NumPy
- Tkinter

Steps Involved in Building the Project

1. Designed the application workflow.
2. Implemented message encryption.
3. Developed image steganography algorithms.
4. Created a graphical user interface.
5. Added message extraction functionality.
6. Integrated encryption and decryption.
7. Tested the application using multiple images.
8. Fixed bugs and optimized performance.

Conclusion

The Secure Image Steganography Tool demonstrates how encryption and steganography can be combined to achieve secure communication. The project provides a practical implementation of data hiding techniques using Python.

Overall Internship Learning

During the internship, I enhanced my practical knowledge in Python programming, cybersecurity, web application security, cryptography, secure software development, and GitHub version control. These projects strengthened my problem-solving skills and provided hands-on experience in developing real-world cybersecurity applications.

Outcome

Both projects were successfully developed, tested, documented, and uploaded to GitHub. They demonstrate practical implementation of cybersecurity concepts and secure software development practices.